

IBM Bob Technology Usage

Project: EcoPFR-AI

AI-Driven Digital Twin and Optimization of Plug Flow Reactors

1. Overview

EcoPFR-AI is an AI-assisted Plug Flow Reactor (PFR) digital-twin platform designed to optimize reactor operation for high conversion, reduced residence time, lower energy consumption, and reduced environmental impact.

The system combines:

- Chemical engineering-based PFR modelling
- Mass and energy balances
- Reaction kinetics
- Residence-time distribution (RTD) analysis
- Machine learning using XGBoost
- Optimization using Differential Evolution
- Renewable-energy/solar integration
- Sustainability and carbon-impact analysis

IBM Bob was used as an AI-powered software development partner during the design and implementation of the EcoPFR-AI software system.

2. Role of IBM Bob

IBM Bob was used throughout the software development lifecycle to assist the team with:

- Project architecture and planning
- Code generation
- Code explanation
- Debugging
- Refactoring
- Documentation
- Testing and validation
- Frontend development
- Backend development
- Integration of different software modules
- Reviewing and improving implementation logic

Bob allowed the team to describe software requirements using natural language and iteratively develop the application while retaining human review and control over the generated changes.

3. Project Architecture Designed with Bob Assistance

The EcoPFR-AI architecture consists of the following major layers:

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